Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customers segments, product preferences and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3900
- Columns: 18
- Key Features:
    - Customer demographics(Age, Gender, Location, Subscription status)
    - Purchase details(Item Purchase, Category, Purchase Amount, Season, Size, Color)
    - Shopping behavior(Discount Applied, Promo Code Use, Previous Purchase, Frequency of Purchase, Review Ratting, Shipping Type)
- Missing data: 37 in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning with python:

- **Data Loading:** Imported dataset using pandas.

```
[4]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
 4   Category                3900 non-null   str
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   str
 7   Size                    3900 non-null   str
 8   Color                   3900 non-null   str
 9   Season                  3900 non-null   str
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str
 12  Shipping Type           3900 non-null   str
 13  Discount Applied        3900 non-null   str
 14  Promo Code Used         3900 non-null   str
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   str
 17  Frequency of Purchases  3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 835.9 KB
```

- **Initial Exploration:** Used df.info() to check structure and df.describe() for summary statistics.

```
df.describe()
```

	customer_id	age	purchase_amount	review_rating	previous_purchases	purchase_frequency_days
count	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750051	25.351538	80.148462
std	1125.977353	15.207589	23.685392	0.713590	14.447125	120.883544
min	1.000000	18.000000	20.000000	2.500000	1.000000	7.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000	14.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000	30.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000	90.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000	365.000000

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

```
[7]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

- **Column Standardization:** Renamed columns snake case for better readability and documentation.
- **Feature Engineering:**
  - Create age_group column by binning customer age

```
bins = [17,25,40,60,100]
labels = ['Young Adult', 'Adult', 'Middle Age', 'Senior']
df['age_group'] = pd.cut(df['age'], bins= bins, labels= labels)
```

  - Create purchase_frequency_days from purchase data.

```
[27]: frequency_mapping = {

    'Fortnightly':14,
    'Weekly': 7,
    'Annually': 365,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Monthly':30,
    'Every 3 Months': 30
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to MySQLWorkbench and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL(Business Transactions)

We performed structured analysis in MySQLWorkbench to answer key business questions.

1. **Revenue by Gender:** Compared total revenue generated by male and female customers.

Result Grid

gender	revenue
Male	157890
Female	75191

2. **High-Spending Discount Users:** Identified customers who used discounts but still spent above the average purchase amount.

Result Grid Filter Rows:

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88

3. **Top 5 products by Rating:** Found products with the highest average review rating.

item_purchas...	Average Product Review
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. **Shipping Type Comparison:** Compared average purchase amounts between Standard and Express Shipping.

Result Grid Filter Rows:

shipping_type	average_revenue
Express	60.48
Standard	58.46

5. **Subscriber Vs. Non-Subscribers:** Compared average spend and total revenue across subscription status.

Result Grid Filter Rows: Search Export:

subscription_status	total_customer	avg_revenue	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. **Discount-Dependent Products:** Identified 5 products with the highest percentage of discounted purchases.

Result Grid Filter Rows: Sear

item_purchased	discount_rate
Coat	49.07
Hat	50.00
Pants	47.37
Sneakers	49.66
Sweater	48.17

7. **Customer Segmentation:** Classified customers into New, Returning and Loyal segment based on purchase history.

Result Grid Filter Rows: Search

customer_segment	Number of Customer
New	83
Returning	701
Loyal	3116

8. **Top 3 products based on Category:** Listed the most purchased products within each category.

Result Grid Filter Rows: Search Export:

category	item_purchased	total_purchased	product_rank
Accessories	Jewelry	171	1
Clothing	Pants	171	1
Footwear	Sandals	160	1
Outerwear	Jacket	163	1
Accessories	Belt	161	2
Clothing	Blouse	171	2
Footwear	Shoes	150	2
Outerwear	Coat	161	2
Accessories	Sunglasses	161	3
Clothing	Shirt	169	3
Footwear	Sneakers	145	3

9. **Repeat Buyers and Subscriptions:** Checked whether customers with >5 purchases more likely to subscribe.

Result Grid Filter Rows:

subscription_status	repeat_buyers
Yes	958
No	2518

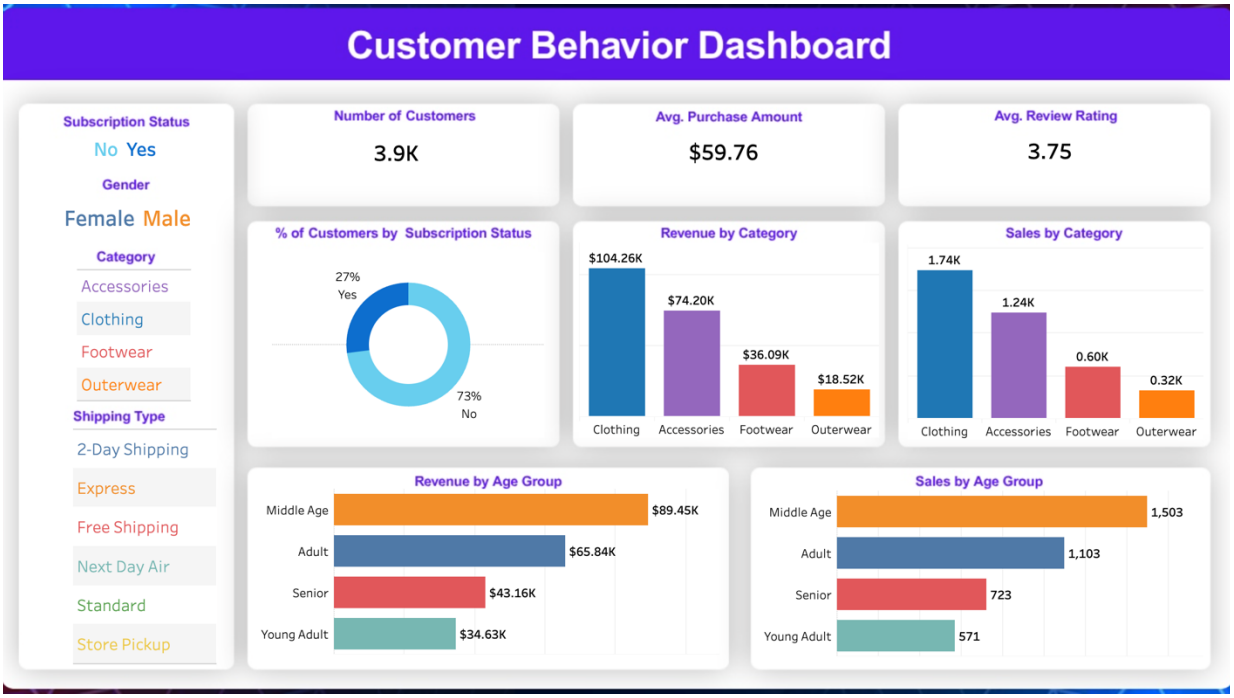
10. **Revenue by Age Group:** Calculated total revenue contribution of each age group.

Result Grid Filter Rows:

age_group	total_revenue
Middle Age	89445
Adult	65842
Senior	43164
Young Adult	34630

5. Dashboard in Tableau

Finally, we build an interactive dashboard in Tableau to present insights visually.



6. Business Recommendations

- Boost Subscriptions: Promote exclusive benefits for subscribers.
- Customer Loyalty Programs: Reward repeat buyers to move them into “Loyal ” segment.
- Review Discount Policy: Balance sales boosts with margin control.
- Product Positioning: Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing: Focus efforts on high-revenue age groups and express-shipping users. Also focus on Young Adult customer give some offers or coupons code only this category of customers.